

Database Management System (DBMS)

Basic Notes

1. What is DBMS?

A **Database Management System (DBMS)** is software used to create, store, organize, retrieve, and manage data efficiently.

Examples

- MySQL
- Oracle
- PostgreSQL
- Microsoft SQL Server
- MongoDB (NoSQL)

Real-Life Example

A college database stores:

- Student details
- Faculty information
- Course records
- Marks
- Attendance

2. Advantages of DBMS

- Reduces data redundancy
- Improves data consistency
- Provides security
- Supports backup and recovery
- Enables data sharing
- Maintains data integrity

3. Disadvantages of DBMS

- Expensive
 - Requires skilled administrators
 - Complex software
 - High hardware requirements
 - Performance overhead
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4. Components of DBMS

- **Hardware** – Computer, storage devices
 - **Software** – DBMS software
 - **Data** – Stored information
 - **Users** – DBA, developers, end users
 - **Procedures** – Rules for using the database
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5. Database Architecture

One-Tier

Database and application are on the same system.

Two-Tier

Client communicates directly with the database.

Three-Tier

Presentation Layer → Application Layer → Database Layer

Most modern applications use **Three-Tier Architecture**.

6. Data Models

Hierarchical Model

Tree structure (Parent → Child)

Network Model

Many-to-many relationships

Relational Model

Stores data in tables.

Example:

Student Table

Roll Name Branch

101 Rahul CSE

102 Priya ECE

This is the most commonly used model.

7. Database Languages

DDL (Data Definition Language)

Used to define database structure.

Commands:

- CREATE
- ALTER
- DROP
- TRUNCATE
- RENAME

Example:

```
CREATE TABLE Student(  
Roll INT,  
Name VARCHAR(30)  
);
```

DML (Data Manipulation Language)

Used to manipulate data.

Commands:

- INSERT

- UPDATE
- DELETE

Example:

```
INSERT INTO Student VALUES(101,'Rahul');
```

DQL (Data Query Language)

Used to retrieve data.

Command:

```
SELECT * FROM Student;
```

DCL (Data Control Language)

Controls permissions.

Commands:

- GRANT
 - REVOKE
-

TCL (Transaction Control Language)

Commands:

- COMMIT
 - ROLLBACK
 - SAVEPOINT
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8. Keys in DBMS

Primary Key

Uniquely identifies each row.

Example:

Student Roll Number

Candidate Key

Possible primary keys.

Alternate Key

Candidate key not selected as primary key.

Foreign Key

Creates relationship between two tables.

Example:

Student

Roll DeptID

101 1

Department

DeptID DeptName

1 CSE

DeptID is the Foreign Key.

Composite Key

Combination of multiple columns.

Super Key

One or more attributes uniquely identify records.

9. Constraints

- NOT NULL
- UNIQUE
- PRIMARY KEY

- FOREIGN KEY
 - CHECK
 - DEFAULT
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10. ER Model

Used for database design.

Components

Entity

Example:

Student

Attribute

Example:

Roll, Name, Age

Relationship

Example:

Student studies Course

11. Normalization

Removes redundancy.

1NF

No repeating groups.

2NF

Removes partial dependency.

3NF

Removes transitive dependency.

BCNF

Stronger version of 3NF.

Benefits:

- Reduces duplication
- Improves consistency

- Saves storage
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12. ACID Properties

Atomicity

Either complete transaction or none.

Consistency

Database remains valid.

Isolation

Transactions execute independently.

Durability

Committed data remains even after power failure.

13. Transaction

A transaction is a logical unit of work.

Example:

Money Transfer

1. Debit Account A
2. Credit Account B

Both operations must complete successfully.

14. SQL Joins

INNER JOIN

Returns matching rows.

LEFT JOIN

Returns all left table records.

RIGHT JOIN

Returns all right table records.

FULL JOIN

Returns all records from both tables.

15. Views

A virtual table created from SQL queries.

Example:

```
CREATE VIEW StudentView AS
```

```
SELECT Name, Branch
```

```
FROM Student;
```

16. Index

Used to speed up searching.

Advantages:

- Faster retrieval
- Improves query performance

Disadvantage:

- Requires extra storage
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17. DBMS vs File System

DBMS	File System
Less redundancy	More redundancy
High security	Low security
Backup available	Limited backup
Concurrent access	Limited access
Supports SQL	No SQL

18. Database Users

- Database Administrator (DBA)

- Application Programmer
 - End User
 - Database Designer
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19. Common SQL Commands

CREATE DATABASE College;

```
CREATE TABLE Student(  
Roll INT PRIMARY KEY,  
Name VARCHAR(30),  
Branch VARCHAR(20)  
);
```

```
INSERT INTO Student VALUES  
(101,'Rahul','CSE');
```

```
SELECT * FROM Student;
```

```
UPDATE Student  
SET Branch='IT'  
WHERE Roll=101;
```

```
DELETE FROM Student  
WHERE Roll=101;
```